

**REG NO: 22TD0005 NAME: AKSHAYA S YEAR/SEM/SEC: Y3/S6/B**

**AIM:**

**SOURCE CODE:**

```
GLOBAL __main
        AREA main, CODE, READONLY
        EXPORT __main
        EXPORT __use_two_region_memory
__use_two_region_memory EQU 0
        BX    LR
        ENTRY

;Main loop begins

__main
        LDR R0,=0X00011111 ;load first input
        LDR R1,=0X54711212 ;load second input
        ADD r2,r0,r1

stop      B stop

        end
```

**RESULT:**

**REG NO: 22TD0005 NAME: AKSHAYA S YEAR/SEM/SEC: Y3/S6/B**

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**SOURCE CODE:**

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GLOBAL __main
        AREA main, CODE, READONLY
        EXPORT __main
        EXPORT __use_two_region_memory
__use_two_region_memory EQU 0
        BX    LR
        ENTRY

;Main loop begins

__main
        LDR R0,=0X00011111 ;load first input
        LDR R1,=0X54711212 ;load second input
        SUB r2,r0,r1

stop      B stop

        end
```

**RESULT:**

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**SOURCE CODE:**

```
GLOBAL __main
        AREA main, CODE, READONLY
        EXPORT __main
        EXPORT __use_two_region_memory
__use_two_region_memory EQU 0
        BX    LR
        ENTRY

;Main loop begins

__main
        LDR R0,=0X00011111 ;load first input
        LDR R1,=0X54711212 ;load second input
        MUL  r2,r0,r1

stop      B stop

        end
```

**RESULT:**

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**SOURCE CODE:**

```
GLOBAL __main
        AREA main, CODE, READONLY
        EXPORT __main
        EXPORT __use_two_region_memory
__use_two_region_memory EQU 0
        BX    LR
        ENTRY

;Main loop begins

__main
        LDR  R0,=0x00000006 ;load first input
        LDR  R1,=0x00000004 ;load second input
        ADD  r2,R0,R1
        SUBS R3,R0,R1
        MUL  R4,R0,R1

stop      B stop

        end
```

**RESULT:**

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**AIM:**

**SOURCE CODE:**

```
GLOBAL __main
        AREA main, CODE, READONLY
        EXPORT __main
        EXPORT __use_two_region_memory
__use_two_region_memory EQU 0
        BX    LR
        ENTRY

;Addresses of Registers
IO0DIR    EQU    0xE0028008
IO1DIR    EQU    0xE0028018
IO0PIN    EQU    0xE0028000
IO1PIN    EQU    0xE0028010
IO0SET    EQU    0xE0028004
IO0CLR    EQU    0xE002800C
IO1CLR    EQU    0xE002801C
IO1SET    EQU    0xE0028014
PINSEL0    EQU    0xE002C000
PINSEL1    EQU    0xE002C004
PINSEL2    EQU    0xE002C014
;

__main

        ldr    r1,=IO1DIR
        ldr    r2,=0x00FF0000    ;initialize input & output
        str    r2, [r1]

loop
        ldr    r1,=IO1PIN
        LDR    r2, [r1]
        LSR    R2,#8
        AND    R2,#0x00ff0000
```

```
str  r2, [r1]
B    loop
END
```

**RESULT:**

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**AIM:**

**SOURCE CODE:**

```
GLOBAL __main
        AREA main, CODE, READONLY
        EXPORT __main
        EXPORT __use_two_region_memory
__use_two_region_memory EQU 0
        BX     LR
        ENTRY

;Addresses of Registers
IO0DIR      EQU      0xE0028008
IO1DIR      EQU      0xE0028018
IO0PIN      EQU      0xE0028000
IO1PIN      EQU      0xE0028010
IO0SET      EQU      0xE0028004
IO0CLR      EQU      0xE002800C
IO1CLR      EQU      0xE002801C
IO1SET      EQU      0xE0028014
PINSEL0     EQU      0xE002C000
PINSEL1     EQU      0xE002C004
PINSEL2     EQU      0xE002C014
LEDDELAY    EQU      800000
;

__main
        ldr     r1,=IO0DIR                ;IO1DIR OUTPUT
CONFIG
        ldr     r2,=0x00FF0000
        str     r2, [r1]

loop
        ldr     r1,=IO0PIN                ;IO1PIN
        ldr     r2,=0x00FF0000
```

```

    str    r2, [r1]
    BL     __delay

    ldr    r1,=IO0PIN
    ldr    r2,=0x00000000
    str    r2, [r1]
    BL     __delay

    B      loop

```

;SOLF DELAY

;Delay loop

;Constant calculations

;1/30Mhz=1/30us=0.03334us

;1us requires 30 counts ie 30x0.0334=1us

;So 1ms requires 30x1000=30000 counts but the SUBS and BNE combination requires 2cy

;So the constant final =30000/2=15000;

\_\_delay

```

    ldr r0,=0x20000

```

doloop

```

    subs r0,#1
    bne doloop
    bx lr

```

END

**RESULT:**



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**SOURCE CODE:**

```
GLOBAL __main
        AREA main, CODE, READONLY
        EXPORT __main
        EXPORT __use_two_region_memory
__use_two_region_memory EQU 0
        BX    LR
        ENTRY
```

;Addresses of Registers

|         |     |            |
|---------|-----|------------|
| IO0DIR  | EQU | 0xE0028008 |
| IO1DIR  | EQU | 0xE0028018 |
| IO0PIN  | EQU | 0xE0028000 |
| IO1PIN  | EQU | 0xE0028010 |
| IO0SET  | EQU | 0xE0028004 |
| IO0CLR  | EQU | 0xE002800C |
| IO1CLR  | EQU | 0xE002801C |
| IO1SET  | EQU | 0xE0028014 |
| PINSEL0 | EQU | 0xE002C000 |
| PINSEL1 | EQU | 0xE002C004 |
| PINSEL2 | EQU | 0xE002C014 |

;

\_\_main

```
ldr    r1,=IO1DIR           ;IO1DIR
ldr    r2,=0x00FF0000
str    r2, [r1]
```

loop

```
ldr    r1,=IO1PIN           ;IO1PIN
ldr    r2,=0x00FF0000
str    r2, [r1]
```

```

BL    __delay

ldr   r1,=IO1PIN
ldr   r2,=0x00000000
str   r2, [r1]
BL    __delay

B     loop

```

;Delay loop  
 ;Constant calculations  
 ; $1/30\text{Mhz}=1/30\mu\text{s}=0.03334\mu\text{s}$   
 ;1us requires 30 counts ie  $30 \times 0.0334=1\mu\text{s}$   
 ;So 1ms requires  $30 \times 1000=30000$  counts but the SUBS and BNE combination requires 2cy  
 ;So the constant final  $=30000/2=15000$ ;

```

__delay
STMDB    sp!, {r14}
LDR      r12, =3000000

LOOP
SUBS    r12, r12, #1
BNE     LOOP
LDMIA   sp!, {r14}
BX      r14

END

```

**RESULT:**